

SEMIPRIME RIGIDITY AND GROWTH ALONG POWERS IN INTEGER-SIDED CYCLIC POLYGONS

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ABSTRACT. For a positive integer n , let $a(n)$ be the largest number of sides of a simple cyclic polygon of circumradius n whose side lengths are positive integers. We classify semiprime radii exactly: $a(pq)$ can only be 6, 8, or 18, with the exceptional values characterized by Gaussian and Eisenstein angle conditions. In sharp contrast, for every fixed integer $N \geq 2$, the values along its powers satisfy $a(N^e) \rightarrow \infty$ and $\limsup_{e \rightarrow \infty} a(N^e)/e \geq 2$. A quadratic chord tower and Farey neighbors give explicit constructions, including $a(8192) \geq 38$. We also prove that $a(n)$ is even whenever at most one distinct prime divisor of n is congruent to 1 (mod 4). The proofs use Conway–Radin–Sadun angle splitting, arithmetic in quadratic fields, an exact finite verification for even semiprime radii, and a one-dimensional Farey bound.

1. INTRODUCTION

A cyclic polygon is a polygon whose vertices lie on a common circle. We fix the circumradius to be a positive integer, require every side length to be a positive integer, and ask how many sides are possible. Classical arithmetic work on cyclic polygons has focused especially on area: Robbins’s formulas and their extensions relate the area to the side lengths [5, 3]. Here the question is different. Which integer chords can close around the circle? The obstruction is the exact additive relation among their half-angles.

For $n \geq 1$ and $1 \leq s \leq 2n$, put

$$(1) \quad \alpha_n(s) = \arcsin\left(\frac{s}{2n}\right) \in (0, \pi/2].$$

Thus $2\alpha_n(s)$ is the minor central angle subtended by a chord of length s .

Definition 1.1. Let $a(n)$ be the largest integer k for which there is a simple cyclic k -gon of circumradius n with positive integer side lengths.

For each fixed n , there are only finitely many candidate side lengths, and every side contributes an arc gap of at least $2\alpha_n(1) > 0$. Thus the number of sides is bounded and the maximum $a(n)$ exists. Polygon closure couples the half-angles through exact relations, so $a(n)$ records how the arithmetic of the radius shapes the supply of compatible chords.

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The function $a(n)$ is OEIS A398490; the corresponding counting triangle is A398489 [4]. The regular hexagon gives $a(n) \geq 6$ for every n . The first surprise is that even among radii with only two prime factors, counted with multiplicity, the answer is not constant. The examples in Table 1 show the three semiprime outcomes and the contrasting growth along powers. The jumps are arithmetic rather than metric. For a prime $r \equiv 1 \pmod{3}$, choose the nontrivial solution

$$(2) \quad s_r^2 + 3u_r^2 = 4r^2, \quad s_r, u_r > 0,$$

with $s_r > 0$ minimal, and put

$$(3) \quad x_r = \arcsin\left(\frac{s_r}{2r}\right) \in (0, \pi/6).$$

Three phenomena set the agenda. Semiprime radii are completely classified, with only three possible maxima. By contrast, increasing exponents at any fixed prime support force unbounded growth. Yet a parity constraint survives for all radii with at most one distinct prime divisor congruent to 1 (mod 4). All three stories begin with the exact angle-closure relation; what changes is how its arithmetic structure constrains, or creates, families of solutions. Farey adjacency is the recurring engine behind the latter two: a Stern–Brocot path forces exact cancellation in the growth construction, while Farey neighbors give the sharp Gaussian rank-one bound behind parity. We begin with the sharpest case, the semiprime classification.

Theorem 1.2 (Semiprime rigidity). *Let $p \leq q$ be primes. Then*

$$a(pq) = \begin{cases} 18, & p, q \text{ are odd, } p \equiv q \equiv 1 \pmod{3}, \\ & x_p + x_q < \pi/6, \\ 8, & p < q, \quad p \equiv q \equiv 1 \pmod{4}, \quad \text{and the first case fails,} \\ 6, & \text{otherwise.} \end{cases}$$

Moreover, $x_p + x_q < \pi/6$ is equivalent to

$$(4) \quad 3(u_p s_q + u_q s_p) < 3u_p u_q - s_p s_q.$$

In particular,

$$a(pq) \in \{6, 8, 18\}, \quad a(2p) = a(3p) = 6$$

for every prime p .

The 6–8–18 rigidity phenomenon. Among all prime pairs $p \leq q$, the semiprime maximum $a(pq)$ takes only the values

$$6, \quad 8, \quad 18.$$

This is a complete classification, including the repeated-prime case $p = q$.

Figure 1 turns the classification into a two-step test. The condition at the first branch is precisely the first case of Theorem 1.2; only when it fails do we test for the Gaussian value.

The semiprime theorem says that low arithmetic complexity imposes strong rigidity. One might therefore expect that fixing the prime support

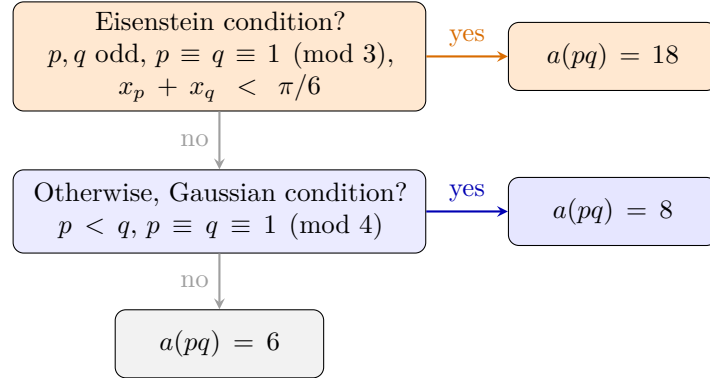


FIGURE 1. A decision diagram for the complete semiprime classification. The Gaussian test is applied only after the Eisenstein condition fails.

keeps $a(n)$ uniformly bounded. The next theorem shows that this intuition is false: increasing exponents alone can produce arbitrarily many sides.

Theorem 1.3 (Growth along powers). *For every integer $N \geq 2$,*

$$a(N^e) \longrightarrow \infty \quad (e \rightarrow \infty),$$

and more quantitatively

$$(5) \quad \limsup_{e \rightarrow \infty} \frac{a(N^e)}{e} \geq 2.$$

The construction behind Theorem 1.3 is elementary once the right angle is chosen. Powers of a single quadratic integer produce integer chords at the radii N^e . A Stern–Brocot path then supplies infinitely many pairs of folded angles whose irrational parts cancel exactly. For $N = 2$, one such pair gives the concrete 38-gon $\{896^{[26]}, 2339^{[12]}\}$ at radius $8192 = 2^{13}$.

Remark 1.4 (What the lower bound says for powers of 2). For $N = 2$, put $\tau = 2 \arg(1 + \sqrt{-15})/\pi$. The odd–odd Farey edges below give explicit lower bounds from the construction in the proof.

Exponent e	Farey edge	$2(b_1 + b_2)$	Consequence
2	$1/2 < \tau < 1/1$	6	$a(2^2) \geq 6$
4	$3/4 < \tau < 1/1$	10	$a(2^4) \geq 10$
6	$5/6 < \tau < 1/1$	14	$a(2^6) \geq 14$
13	$5/6 < \tau < 11/13$	38	$a(2^{13}) \geq 38$

These are construction bounds, not exact values of $a(2^e)$. The factor 2 in Theorem 1.3 comes from doubling an exact H -block; the paper gives no matching upper bound, so it does not decide whether the true growth is linear or faster.

There is nevertheless a residual rigidity that survives this growth. Let

$$\omega_1(n) = \#\{q : q \mid n, \text{ } q \text{ prime, } q \equiv 1 \pmod{4}\},$$

where distinct primes are counted once.

Theorem 1.5 (Parity in Gaussian rank one). *If $\omega_1(n) \leq 1$, then $a(n)$ is even. In particular,*

$$a(p^e) \equiv 0 \pmod{2}$$

for every prime p and every $e \geq 1$.

The central contrast. Semiprime radii allow only 6, 8, or 18 sides. Along every fixed-base power sequence N^e , the maximum tends to infinity. Yet when at most one distinct prime divisor of n is congruent to 1 (mod 4), the maximum is still even.

Radius n	$a(n)$	Illustrative regime
49	18	Eisenstein case at a prime square
65	8	Gaussian semiprime case
91	18	Eisenstein semiprime case
361	6	Semiprime baseline
8192	≥ 38	Growth along powers of 2

TABLE 1. Small examples illustrating the three main results.

The plotted values in Figure 2 give a computational view of the first part of $a(n)$. Semiprime radii are highlighted because their values are proved by Theorem 1.2; the remaining points are included as context from the OEIS tabulation, not as claims established here.

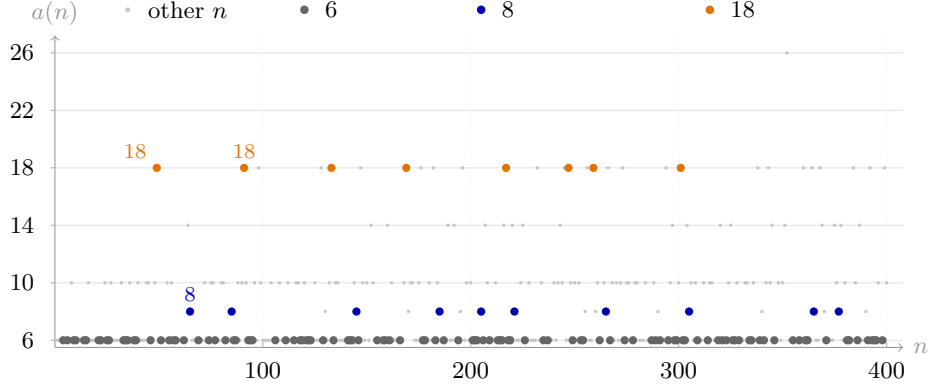


FIGURE 2. OEIS A398490 for $n \leq 400$ [4]. Light gray: other radii; dark gray: semiprime value 6; blue: Gaussian value 8; orange: Eisenstein value 18 (selected points are labeled). The theorem classifies semiprimality; all other points are included for illustration only, and no claims about their plotted values are proved here.

At radius 8192, just two integer side lengths already close a 38-gon: 896 occurs 26 times and 2339 occurs 12 times. Figure 3 previews the polygon and the Farey-neighbor cancellation that makes this exact closure possible.

The exact angle-closure relation branches into three arithmetic stories: Conway–Radin–Sadun splitting yields semiprime rigidity; a quadratic chord

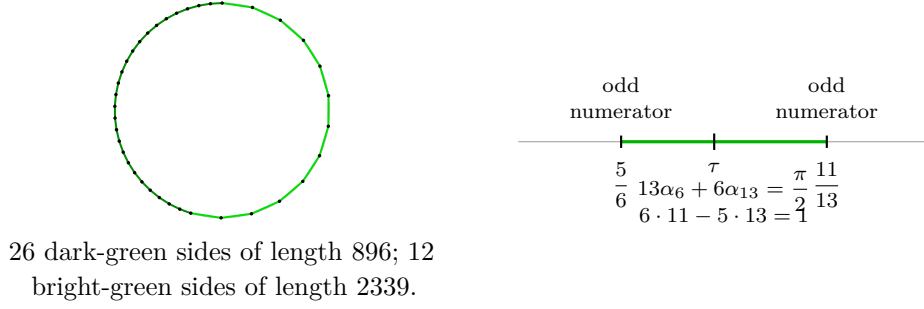


FIGURE 3. The 38-gon at radius 8192 and its Farey construction. Its two side lengths give the exact closure relation $26\alpha_6 + 12\alpha_{13} = \pi$; the right panel shows the determinant-one bracket behind it.

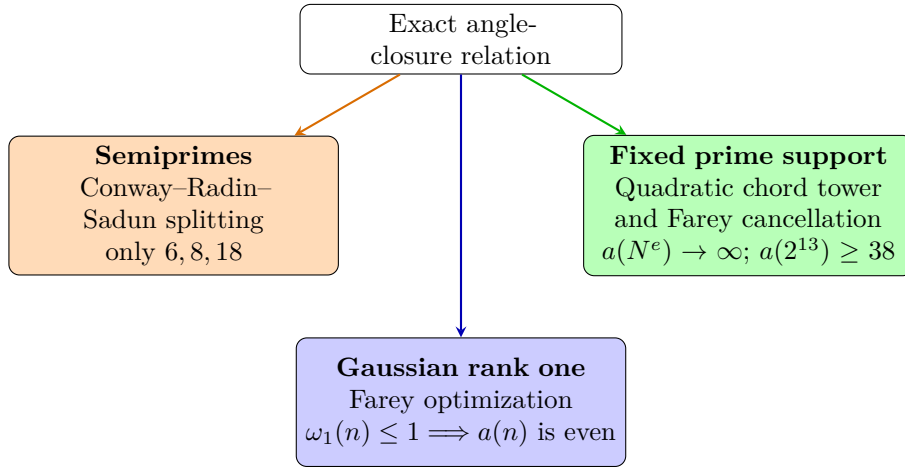


FIGURE 4. Rigidity and growth under the same exact closure condition: semiprimes allow only 6, 8, 18, fixed-support power sequences are unbounded (already $a(2^{13}) \geq 38$), and Gaussian rank one retains evenness.

tower with Farey cancellation gives unbounded growth; and Farey optimization in Gaussian rank one yields parity.

The figures use a semantic color key: blue marks Gaussian and parity structures, orange marks Eisenstein and 18-gon structures, green marks growth and Farey constructions, and gray marks general geometry. In the initial-value plot, dark gray distinguishes the semiprime value 6, while blue and orange mark the Gaussian value 8 and Eisenstein value 18.

2. THE GEOMETRIC CRITERION

Lemma 2.1 (Minor- and major-arc criterion). *A multiset of positive integers $\{s_1, \dots, s_k\}$, with $s_i \leq 2n$, is realizable as the side-length multiset of a simple cyclic polygon of radius n if and only if either*

$$(6) \quad \sum_{i=1}^k \alpha_n(s_i) = \pi,$$

or

$$(7) \quad \alpha_n(s_j) = \sum_{i \neq j} \alpha_n(s_i)$$

for some j .

Proof. If every side uses its minor arc, the central angles $2\alpha_n(s_i)$ sum to 2π , giving (6). A simple cyclic polygon can use a major arc for at most one side. If s_j uses it, its central angle is $2\pi - 2\alpha_n(s_j)$, while all other central angles are minor; their sum is 2π exactly when (7) holds. Conversely, arranging the corresponding arcs consecutively around the circle constructs a simple cyclic polygon. $\square$

Figure 5 depicts the two possibilities in the criterion. The angles $\alpha_n(s_i)$ are half of the corresponding central angles.

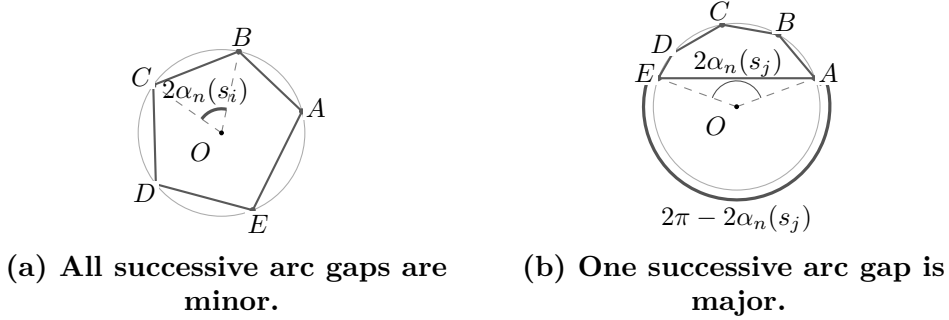


FIGURE 5. The two arc configurations behind Lemma 2.1. In (b), the dark-gray outer arc is the major gap; its chord has minor central angle $2\alpha_n(s_j)$.

We call (6) a π -block. A positive relation with sum $\pi/2$ is called a $\pi/2$ -block. Relation (7) is a major-arc relation. A side of length $2n$ is a diameter; in this case the remaining sides form a $\pi/2$ -block.

We shall also use the strict monotonicity

$$(8) \quad \frac{d}{ds} \alpha_n(s) = \frac{1}{\sqrt{4n^2 - s^2}} > 0.$$

Thus a half-angle uniquely determines its side length at fixed radius.

Proposition 2.2 (Scaling). *For all positive integers m, n ,*

$$a(mn) \geq a(n).$$

Proof. Scale every side length of a polygon of radius n by m . All central angles are unchanged, and the new polygon has radius mn . $\square$

3. GEODETIC ANGLES AND COMPONENT CANCELLATION

For a non-diameter side s , write

$$(9) \quad 4n^2 - s^2 = du^2,$$

where d is squarefree. Its complementary half-angle

$$\beta_n(s) = \frac{\pi}{2} - \alpha_n(s)$$

satisfies

$$(10) \quad \tan \beta_n(s) = \frac{u}{s} \sqrt{d}.$$

Hence $\beta_n(s)$ is a pure geodetic angle in the terminology of Conway, Radin, and Sadun [1].

We use their basis theorem and splitting theorem. In a fixed squarefree class d , write the unique basis expansion of a pure geodetic angle as

$$(11) \quad \theta = q(\theta)\pi + \sum_r c_r(\theta)\Theta_{r,d},$$

where the sum is finite and the prime-indexed basis angles $\Theta_{r,d}$ are independent modulo rational multiples of π .

Lemma 3.1 (Component cancellation). *Fix d . If*

$$\sum_i m_i \theta_i \in \pi \mathbb{Q}$$

for pure geodetic angles θ_i in class d and integers m_i , then

$$\sum_i m_i c_r(\theta_i) = 0$$

for every prime-basis element $\Theta_{r,d}$ in (11). Consequently, the signed components belonging to distinct split rational primes cancel separately.

Proof. Insert the unique expansions (11). After subtracting the rational multiple of π , uniqueness of the Conway–Radin–Sadun basis forces the coefficient of every $\Theta_{r,d}$ to vanish. $\square$

The splitting theorem further says that, in any rational relation among pure geodetic angles, the restriction to each squarefree class d is itself a rational multiple of π . These two facts will be used throughout without further comment.

The following parity principle is a useful first consequence. Besides proving the odd-sided criterion, it will later isolate the genuinely difficult part of the parity problem.

Lemma 3.2 (Non-Gaussian parity principle). *Let $\alpha_1, \dots, \alpha_m$ be half-angles of non-diameter integer chords of the same radius, none belonging to the Gaussian class $d = 1$. Let $\varepsilon_i \in \{1, -1\}$, and suppose*

$$\sum_{i=1}^m \varepsilon_i \alpha_i = tH, \quad H = \frac{\pi}{2}, \quad t \in \mathbb{Z}.$$

Then

$$\sum_{i=1}^m \varepsilon_i \equiv t \pmod{2}.$$

Proof. Group the terms by their squarefree class $d > 1$. For a side in class d , write

$$4n^2 - s^2 = du^2, \quad \xi = s + u\sqrt{-d}.$$

Then $|\xi| = 2n$ and $\arg \xi = \beta = H - \alpha$. For a fixed class put

$$m_d = \sum_{i \in d} \varepsilon_i, \quad A_d = \sum_{i \in d} \varepsilon_i \alpha_i, \quad B_d = \sum_{i \in d} \varepsilon_i \beta_i = m_d H - A_d.$$

By the splitting theorem, B_d is a rational multiple of π . On the other hand,

$$\prod_{i \in d} \xi_i^{\varepsilon_i} \in \mathbb{Q}(\sqrt{-d})^\times$$

has rational absolute value $(2n)^{m_d}$ and argument B_d . Hence its direction is a root of unity in $\mathbb{Q}(\sqrt{-d})$. For $d \neq 3$ the only roots of unity are ± 1 , so $B_d \in \pi\mathbb{Z}$ and therefore

$$\frac{A_d}{H} \equiv m_d \pmod{2}.$$

For $d = 3$, the roots of unity are the sixth roots, so initially $B_3 \in (\pi/3)\mathbb{Z}$. All classes $d \neq 3$ already have $A_d/H \in \mathbb{Z}$, and the total $\sum_d A_d/H = t$ is an integer. Thus A_3/H is an integer as well. Writing $B_3 = r\pi/3$ gives $B_3/H = 2r/3 \in \mathbb{Z}$, so $3 \mid r$ and in fact B_3/H is even. Hence $A_3/H \equiv m_3 \pmod{2}$ also in the Eisenstein class. Summing over d gives the claim. $\square$

The next theorem completely characterizes when odd side counts are possible: an odd-sided polygon exists exactly when an odd-sided triangle exists, and this happens exactly when the radius has a prime divisor congruent to 1 (mod 4).

Theorem 3.3 (Odd-sided polygons). *For a positive integer n , the following are equivalent:*

- (i) *there exists an odd-sided integer-sided cyclic polygon of radius n ;*
- (ii) *there exists such a triangle;*
- (iii) *n has a prime divisor congruent to 1 (mod 4).*

Proof. The implication (ii) $\Rightarrow$ (i) is immediate. Condition (iii) is equivalent to the existence of a nondegenerate integer right triangle with hypotenuse $2n$; inscribing it in its circumcircle gives (iii) $\Rightarrow$ (ii).

For the converse we argue contrapositively. Suppose that n has no prime divisor congruent to 1 (mod 4). Then there is no non-diameter Gaussian side: such a side would give a nontrivial identity

$$s^2 + u^2 = (2n)^2,$$

and the sum-of-two-squares criterion would force a prime $q \equiv 1 \pmod{4}$ to divide n .

Now consider an admissible polygon with k sides. In the minor-arc case, Lemma 3.2 with all signs positive and $t = 2$ gives $k \equiv 0 \pmod{2}$. For a non-diameter major-arc relation, give the target sign +1 and every source side sign -1. The signed angle sum is zero, while the signed number of sides is $1 - (k - 1) = 2 - k$; the lemma again gives k even. If the target is a diameter, the remaining $k - 1$ half-angles sum to H . Applying the lemma with $t = 1$ shows that $k - 1$ is odd, hence k is even. Thus no odd-sided polygon exists, proving (i) $\Rightarrow$ (iii). $\square$

For later normalization, put

$$(12) \quad g = \gcd(s, u), \quad A = \frac{s}{g}, \quad B = \frac{u}{g}, \quad M = \frac{2n}{g}.$$

Then

$$(13) \quad \gcd(A, B) = 1, \quad A^2 + dB^2 = M^2.$$

Only primes dividing M may occur as prime components of the angle.

4. SEMIPRIME RIGIDITY

The regular hexagon gives the universal baseline 6. To exceed it at a semiprime radius, the angle relation must exploit arithmetic splitting. The Conway–Radin–Sadun theorem separates a closure relation into squarefree classes. Ordinary classes stay within the hexagonal bound; the Gaussian and Eisenstein classes are the exceptions, allowing 8 and 18 sides, respectively. The Eisenstein value occurs exactly under the criterion in Theorem 1.2. The proof below follows this route: first reduce relations to small zero-sum atoms, then classify radii pq with both primes odd structurally, and finally verify the remaining $n = 2p$ bounds over a finite list of ideal classes. Appendix A gives the certificates and independent checks for this last step.

Strategy of the semiprime proof.

- (1) Split each angle relation into minimal zero-sum component atoms.
- (2) Remove the variable prime factors and bound the residual norms.
- (3) Enumerate the finite squarefree classes that can remain.
- (4) Bound side density in the ordinary, Gaussian, and Eisenstein classes.
- (5) Control mixtures and major arcs, then check the remaining $n = 2p$ cases.

4.1. Prime-component atoms. Assume first that $n = pq$ with distinct odd primes p and q .

Lemma 4.1. *If a variable prime $r \in \{p, q\}$ is inert or ramified in $K = \mathbb{Q}(\sqrt{-d})$, then $r \nmid M$ in (13).*

Proof. If r is inert and $r \mid M$, the prime ideal (r) divides $A + B\sqrt{-d}$, and hence $r \mid A, B$, contrary to coprimality. If r is ramified and $r \mid M$, then $r \mid d$. If $r \nmid A$, the valuation of $A^2 + dB^2$ at r is zero; if $r \mid A$, then $r \nmid B$ and squarefreeness of d gives valuation one. Both alternatives contradict the even valuation in M^2 . $\square$

For a split prime r , write $(r) = r\bar{r}$. If $r \mid M$, the signed exponent difference, divided by two, belongs to $\{-1, 0, 1\}$. Thus every primitive side has a normalized variable-component vector

$$(v_p, v_q) \in \{-1, 0, 1\}^2.$$

Every rational angle relation has total component vector zero by Lemma 3.1.

Thus every rational relation gives a zero-sum multiset in the component lattice. The next lemma bounds the size of its minimal pieces; the residual-norm argument then identifies which squarefree classes can occur.

Lemma 4.2 (Minimal component atoms). *Every finite zero-sum multiset of vectors in $\{-1, 0, 1\}^2$ is a disjoint union of minimal zero-sum multisets, and every minimal nonzero one has length at most 4.*

For a repeated prime $p = q$, the corresponding one-dimensional component set is contained in $\{-2, -1, 0, 1, 2\}$, and every minimal nonzero zero-sum multiset has length at most 3.

Proof. Repeatedly remove a minimal nonempty zero-sum submultiset. It remains to bound the length of one minimal atom J of nonzero vectors in

$$V = \{-1, 0, 1\}^2 \setminus \{(0, 0)\}.$$

If J contains opposite vectors v and $-v$, this pair is already a zero-sum submultiset, so minimality forces $J = \{v, -v\}$.

Assume that no opposite pair occurs. Group equal vectors and write the resulting support-minimal positive relation as

$$a_1 v_1 + \cdots + a_r v_r = 0, \quad a_i \in \mathbb{Z}_{>0},$$

with distinct directions v_i . The origin lies in the convex hull

$$0 \in \text{conv}\{v_1, \dots, v_r\}.$$

By Caratheodory's theorem in dimension two, it lies in the convex hull of at most three of these directions. Support minimality therefore gives $r \leq 3$. The cases $r = 1$ and $r = 2$ are impossible without a zero vector or an opposite pair, so $r = 3$.

For three noncollinear vectors u, v, w surrounding the origin, the primitive positive relation is obtained from

$$\det(v, w)u + \det(w, u)v + \det(u, v)w = 0$$

by choosing a common orientation, taking absolute values, and dividing by the common gcd. Every determinant has absolute value 1 or 2. A determinant has absolute value 2 only for two diagonal vectors. Two of the three determinants cannot both have absolute value 2: after fixing their common diagonal vector, the other two diagonal vectors are opposites, which has been excluded. Hence at most one primitive coefficient is 2, and the other two are at most 1. The coefficient sum, which is the length of J , is therefore at most $2 + 1 + 1 = 4$.

In one dimension, apart from opposite pairs, the only primitive minimal relations are $2 - 1 - 1 = 0$ and its negative, so the repeated-prime bound is 3. $\square$

The length-four bound is sharp: the relation

$$2(1, 0) + (-1, 1) + (-1, -1) = 0$$

is a minimal atom, with one direction occurring twice and the other two completing the cancellation.

We now make the residual-norm step completely explicit. Let $R = \mathbb{Z}[\sqrt{-d}]$ and $\mathcal{O}_K$ be the ring of integers of $K = \mathbb{Q}(\sqrt{-d})$.

Lemma 4.3 (Odd division in the quadratic order). *For every odd positive integer m ,*

$$(14) \quad m\mathcal{O}_K \cap R = mR.$$

Proof. If $\mathcal{O}_K = R$, there is nothing to prove. Otherwise $d \equiv 3 \pmod{4}$ and

$$\mathcal{O}_K = \mathbb{Z}\left[\frac{1 + \sqrt{-d}}{2}\right], \quad \mathcal{O}_K/R \cong \mathbb{Z}/2\mathbb{Z}.$$

Let $z \in m\mathcal{O}_K \cap R$, so $z = m\eta$ for some $\eta \in \mathcal{O}_K$. In the quotient $\mathcal{O}_K/R$,

$$0 = [z] = m[\eta].$$

Multiplication by the odd integer m is an automorphism of the group of order two. Hence $[\eta] = 0$, so $\eta \in R$ and $z \in mR$. The reverse inclusion is immediate. $\square$

Lemma 4.4 (Small residual norm). *Let J be a minimal component atom of length $m \leq 4$ for distinct odd primes p, q . Multiply the primitive side elements belonging to J and cancel the equal prime-ideal exponents above p and q . The resulting element $\xi \in R$ satisfies*

$$(15) \quad N(\xi) = 2^{2k} \quad \text{for some } 0 \leq k \leq m \leq 4.$$

In the repeated-prime case the same conclusion holds with $m \leq 3$.

Proof. For each side let $z_j = A_j + B_j\sqrt{-d} \in R$, with $N(z_j) = M_j^2$ and $M_j \mid 2pq$. Put $z = \prod_{j \in J} z_j \in R$. Component cancellation says that the valuations of the two conjugate prime ideals above p are equal, say A , and likewise those above q are equal, say B . Hence

$$(z) = (p)^A (q)^B \mathfrak{a}$$

for an integral ideal $\mathfrak{a}$, and therefore

$$z = p^A q^B \eta$$

for some $\eta \in \mathcal{O}_K$. Since $p^A q^B$ is odd and $z \in R$, Lemma 4.3 gives $\eta \in R$.

After removing the p - and q -parts, only prime factors above 2 remain. Each $M_j \mid 2pq$ contains at most one factor 2, so

$$N(\eta) = 2^{2k}$$

with $k \leq |J| = m$. Renaming η as ξ proves the claim.

If $p = q$, the same argument removes the equal conjugate prime-ideal exponents above the single odd prime p . Now $M_j \mid 2p^2$, so each side contributes at most one factor of 2 to the residual norm. The repeated-prime atom bound $m \leq 3$ therefore gives $k \leq m \leq 3$. $\square$

For the even semiprime radius $n = 2p$, there is only one variable odd prime. If p splits in the class under consideration, its nonzero component vectors are -1 and 1 , so a nonzero minimal component atom has length $m = 2$. Each primitive side norm parameter satisfies $M_j \mid 4p$ and contributes at most two factors of 2. After the p -parts are cancelled, the residual norm is therefore 2^{2k} with $k \leq 2m = 4$. A zero-component side forms a singleton atom. If p splits and $p \mid M_j$, zero component forces equal positive exponents at the two primes above p , so (p) divides the side element in $\mathcal{O}_K$; Lemma 4.3 and primitivity then rule this out. If p is inert or ramified, the argument of Lemma 4.1 rules out $p \mid M_j$. Thus $p \nmid M_j$, so $M_j \mid 4$ and its own residual exponent has $k \leq 2$. Thus the candidate equation (16) with $L \leq 16$ also covers $n = 2p$.

If the residual element is nonreal, divide its two coefficients by their gcd. Since its norm is supported only at 2, this gives a primitive positive solution

$$(16) \quad x^2 + dy^2 = L^2, \quad L \in \{1, 2, 4, 8, 16\}.$$

The full candidate lists, including repeated-prime and $n = 2p$ cases, appear in Appendix A.1. Their recovery is independently checked by the Maple and Python certificates in Appendix A.

Reduction complete. Every semiprime relation is built from component atoms of size at most four, and only finitely many squarefree classes remain. What follows is a density analysis inside those classes.

For the bounds below, L_θ denotes the largest number of sides in a positive block whose half-angles sum to θ , among the squarefree classes currently under consideration. We write L_{major} for the largest number of sides in a corresponding major-arc relation.

4.2. The ordinary squarefree classes. Every d in (A.1) other than 3 satisfies $d \equiv 7 \pmod{8}$. For odd n , equation (9) then forces s and u to be even. After division by 4, the primitive norm is odd, so the individual angle has no prime-2 basis component. The same conclusion holds for any ordinary class $d \neq 1, 3$: if a nonzero prime-2 component existed, Lemma 4.4 would place d in the finite list.

The only roots of unity in $\mathbb{Q}(\sqrt{-d})$ for $d \neq 1, 3$ are ± 1 . Consequently, a rational restriction in an ordinary class is an integral multiple of

$$H = \frac{\pi}{2}.$$

A minimal atom with $k \leq 4$ sides has original-angle sum $kH - m\pi$. Positivity leaves only

$$(17) \quad k = 3, \quad \sum \alpha = H, \quad \text{or} \quad k = 4, \quad \sum \alpha = \pi.$$

Thus every ordinary class has density at most three sides per H , or one side per unit $U = \pi/6$.

This bound also covers the repeated-prime case. By Lemma 4.2, a minimal ordinary component atom then has at most three sides. Its complementary-angle sum is a positive integral multiple of π and is strictly less than $3H = 3\pi/2$; hence it equals π . The corresponding half-angle sum is therefore H , so the same three-sides-per- H bound holds.

For completeness we record the sharper common four-angle model used for major arcs. Up to conjugation, the nonzero exponent pairs are $(1, 0)$, $(0, 1)$, $(1, 1)$, and $(1, -1)$. Choose their two independent components so that $0 < X \leq Y < H$. The possible non-diameter half-angles are

$$(18) \quad H - X, \quad H - Y, \quad H - Y + X, \quad |H - X - Y|.$$

The boundary $X + Y = H$ is excluded by component cancellation: X and Y have nonzero components at distinct split-prime basis elements. If $X + Y < H$, the coefficient equations show that no nonempty positive rational block exists. If $X + Y > H$, write the multiplicities in the order (18) as $m_1, \dots, m_4$. Component cancellation gives

$$m_1 = m_3 + m_4, \quad m_2 = m_4 - m_3, \quad 0 \leq m_3 \leq m_4.$$

The number of angles is $m_3 + 3m_4$, while their rational sum in units of H is $m_3 + m_4$. Hence

$$(19) \quad L_H \leq 3, \quad L_\pi \leq 6.$$

Writing an angle as $cH + aX + bY$, the functional

$$3c + 2a + 2b$$

is at least 1 on all four active vectors and at most 3 on a target vector. Therefore an ordinary major-arc relation has at most four sides.

4.3. The Gaussian and Eisenstein classes.

4.3.1. *The Gaussian class $d = 1$.* Put $H = \pi/2$. Suppose first that $p < q$ and both primes are congruent to 1 (mod 4). Let x, y be their two independent small Gaussian angles, ordered so that $0 < y < x < \pi/4$.

There are eight nonzero variable-component pairs in $\{-1, 0, 1\}^2$. Gaussian multiplication realizes these pairs, while multiplication by the four units changes only the rational H -part. Folding the resulting arguments into $(0, H]$ gives the eight non-diameter angles shown in Figure 6.

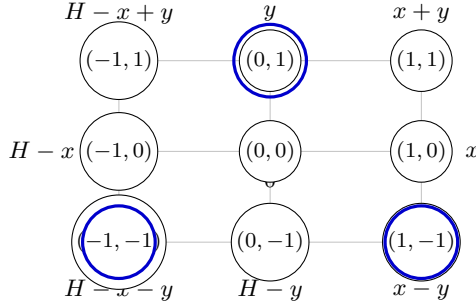


FIGURE 6. The Gaussian coefficient lattice. Horizontal and vertical positions are a and b ; each nonzero pair maps to its folded half-angle $cH + ax + by$. Blue rings mark the three angle types in the octagon construction (22); the angle y is used twice, so they account for four angle terms. Doubling the resulting H -block gives the octagon.

Represent an angle by (c, a, b) if it equals $cH + ax + by$. Every vector in this list satisfies

$$(20) \quad 4c + 2a + b \geq 1.$$

In a rational block, the total x - and y -coefficients vanish. Therefore a sum tH containing L sides satisfies $L \leq 4t$, and in particular

$$(21) \quad L_\pi \leq 8.$$

The bound is attained because

$$(22) \quad (x - y) + y + y + (H - x - y) = H.$$

Doubling gives an octagon.

If

$$s_p^2 + u_p^2 = 4p^2, \quad s_q^2 + u_q^2 = 4q^2,$$

are oriented so that $x > y$, the corresponding side multiset at radius pq is

$$(23) \quad \left\{ \left(\frac{|s_p u_q - u_p s_q|}{2} \right)^{[2]}, (ps_q)^{[4]}, \left(\frac{u_p u_q - s_p s_q}{2} \right)^{[2]} \right\}.$$

Gaussian multiplication proves integrality. For $p = q$, the possible one-variable coefficient pairs are $(2, 0)$, $(-1, 1)$, $(1, 0)$, and $(-2, 1)$ for angles $ax + cH$. Each satisfies $a + 3c \geq 1$, so a π -block has at most six sides. If one variable prime does not split, only a subset of the one-component types occurs and the same bound holds.

For a major-arc relation in the two-variable Gaussian class, the same functional is at most 3 on the target vector and at least 1 on every source vector. Thus there are at most three source sides, and at most four sides including the target. In the one-variable specialization the target functional is at most 2, so its major-arc bound is no larger.

4.3.2. *The Eisenstein class $d = 3$.* Put $U = \pi/6$. Suppose first that $p < q$, both primes are congruent to 1 (mod 3), and write $x = x_p$, $y = x_q$. The boundary $x + y = U$ cannot occur: the two angles have nonzero components at the distinct prime-basis elements indexed by p and q , so their sum is not a rational multiple of π .

The Eisenstein integers have six roots of unity, compared with four in the Gaussian integers and two in the ordinary imaginary quadratic fields. Their unit action creates a richer list of folded angles. In the active chamber $x + y < U$, that list has density 3 sides per unit U ; because a polygon closes with total half-angle $\pi = 6U$, the bound reaches 18. Multiplying the two split prime factors by the six units and folding their arguments into $(0, H]$ gives the coefficient triples (c, a, b) for angles $cU + ax + by$. The list changes when $x + y$ crosses U , as Figure 7 shows.

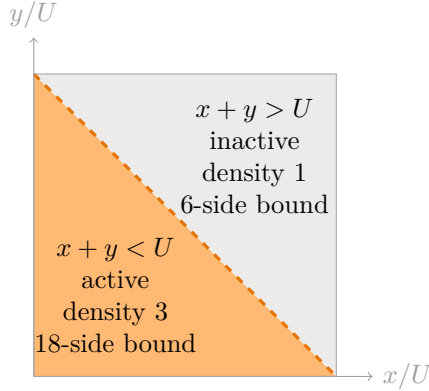


FIGURE 7. The Eisenstein chamber boundary separating the active and inactive angle lists. The dashed diagonal is $x + y = U$; coordinates are normalized by $U = \pi/6$, and density is measured in sides per unit U .

If $x + y < U$, the complete non-diameter coefficient list is

$$\begin{aligned} \mathcal{E}_+ = \{ & (0, 0, 1), (0, 1, 0), (1, -1, -1), (1, -1, 1), (1, 0, 0), \\ & (1, 1, -1), (1, 1, 1), (2, -1, 0), (2, 0, -1), (2, 0, 1), \\ & (2, 1, 0), (3, -1, -1), (3, -1, 1) \}. \end{aligned}$$

The list follows by multiplying the two Eisenstein prime factors, applying the six units, and folding the arguments into $(0, H]$. Every vector satisfies

$$(24) \quad 3c + a + b \geq 1.$$

Thus a sum tU has at most $3t$ sides, and a π -block has at most 18 sides. Equality is attained by taking six copies of

$$(25) \quad x + y + (U - x - y) = U.$$

The third side length is

$$(26) \quad S = \frac{3u_p u_q - s_p s_q - 3(u_p s_q + u_q s_p)}{4}.$$

The numerator is divisible by 4 by Eisenstein multiplication, and $S > 0$ is exactly criterion (4). The resulting side multiset is

$$(27) \quad \{(qs_p)^{[6]}, (ps_q)^{[6]}, S^{[6]}\}.$$

Figure 8 displays the three chord families in the 18-gon and compares the Gaussian and Eisenstein density bounds.

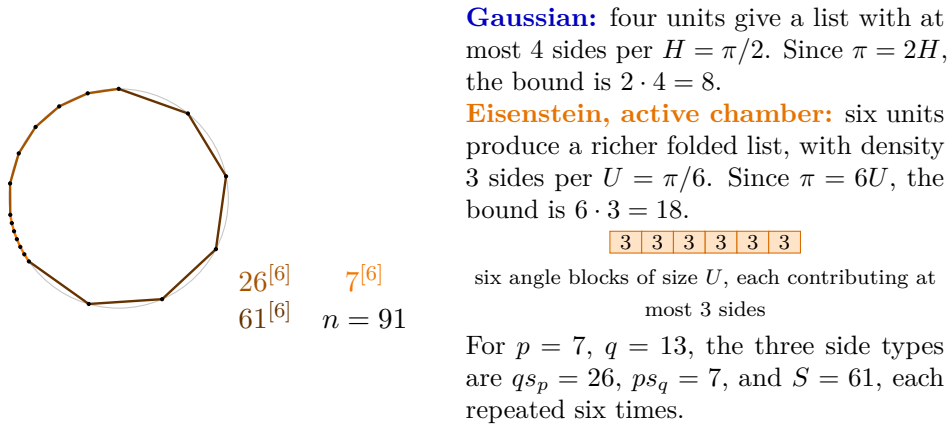


FIGURE 8. Anatomy of the 18-gon at radius 91 (left) and the Gaussian–Eisenstein density comparison (right). The colored blocks represent six sides each; the schematic circle uses the corresponding half-angles. The unit counts describe the folding actions, while the proven density bounds give the numerical maxima.

If $x + y > U$, the complete folded list is

$$\begin{aligned} \mathcal{E}_- = \{ & (-1, 1, 1), (0, 1, 0), (0, 0, 1), (1, 1, -1), (1, 0, 0), \\ & (1, -1, 1), (2, 0, -1), (2, -1, 0), (3, -1, -1), (1, 1, 1), \\ & (2, 1, 0), (2, 0, 1), (3, 1, -1) \}. \end{aligned}$$

Every vector satisfies

$$(28) \quad c + a + b \geq 1.$$

Hence a π -block has at most six sides. If one of p, q does not split in the Eisenstein integers, the available vectors form a subset of the inactive table. For $p = q$, the one-variable specialization gives the same bounds: 18 exactly when $2x_p < U$, and otherwise 6. The boundary $2x_p = U$ is impossible because $\sin^2 x_p$ is rational, whereas $\sin^2(\pi/12) = (2 - \sqrt{3})/4$ is irrational.

Finally, a direct sine expansion gives

$$\sin(U - x_p - x_q) = \frac{3u_p u_q - s_p s_q - 3(u_p s_q + u_q s_p)}{8pq},$$

which proves the equivalence between $x_p + x_q < U$ and (4).

Where we are. Each surviving squarefree class now has its own density bound. The final step is to show that mixing classes cannot surpass the densest one.

4.4. Mixtures and major-arc relations. The rational restrictions in an ordinary or Gaussian class are integral multiples of $H = \pi/2 = 3U$. In the Eisenstein class they are integral multiples of $U = \pi/6$.

Key density principle. A positive mixture of squarefree classes cannot have greater side density than its densest participating class. The global extremal problem therefore reduces to comparing the local densities.

The previous subsections give the following maximal densities, measured in sides per unit U :

$$(29) \quad \rho_d \leq \begin{cases} 3, & d = 3 \text{ and } x_p + x_q < U, \\ 4/3, & d = 1 \text{ with two distinct split variable primes,} \\ 1, & \text{otherwise.} \end{cases}$$

Indeed, neutral atoms with no variable component have density at most one side per unit U .

Here a positive block means a finite multiset of sides whose half-angles sum to a positive rational multiple of π . The splitting theorem shows that restricting such a block to one squarefree class again gives a block of this kind (unless the restriction is empty).

Proposition 4.5 (Linear density under positive mixtures). *Let $U = \pi/6$. Suppose each squarefree class d has a bound ρ_d such that every positive block contained in that class, with angle sum A and k sides, satisfies $k \leq \rho_d A/U$. If a positive block mixes classes and its restriction to class d contributes angle A_d and k_d sides, then*

$$(30) \quad \sum_d k_d \leq \left(\max_d \rho_d \right) \frac{\sum_d A_d}{U}.$$

In particular, a mixed π -block has at most $6 \max_d \rho_d$ sides.

Proof. The splitting theorem makes the restriction to each class a rational multiple of π , so each class contribution is a positive rational block to which the assumed local bound applies. Thus

$$\sum_d k_d \leq \sum_d \rho_d \frac{A_d}{U} \leq \left(\max_d \rho_d \right) \frac{\sum_d A_d}{U}.$$

For a π -block, $\sum_d A_d = \pi = 6U$, which gives the final assertion. $\square$

By (29) and Proposition 4.5, the global maximum is 18 in the active Eisenstein case, 8 in the remaining two-component Gaussian case, and 6 otherwise. The constructions (22) and (25), together with the regular hexagon, attain these bounds.

Lemma 4.6 (No mixed non-diameter major arc). *Every non-diameter major-arc relation is contained in one squarefree class. A diameter target reduces to a $\pi/2$ -block.*

Proof. Write a non-diameter relation as

$$\alpha_* = \sum_j \alpha_j < H,$$

and let the target angle lie in class d_0 . If a non-target class occurs, the restriction to d_0 equals the sum of the positive contributions of all non-target classes, and is therefore positive and strictly smaller than H . Every restriction is rational by the splitting theorem.

If $d_0 \neq 3$, its restriction is an integral multiple of H , impossible in $(0, H)$. If $d_0 = 3$, its restriction is U or $2U$, while every non-target ordinary or Gaussian restriction is a multiple of $H = 3U$; their sum cannot equal U or $2U$. Hence no non-target class occurs. If the target is a diameter, the remaining angles sum to H . $\square$

Inside one class, apply the same functional to the target vector and to the sum of the source vectors. Component cancellation makes the functional additive across the relation. Each source vector contributes at least 1. The target functional is at most 3 in an ordinary or Gaussian class, at most 9 in the active Eisenstein class, and at most 3 in the inactive Eisenstein class, as is seen from the displayed coefficient lists. Thus the major-arc bounds, including the target, are respectively 4, 4, 10, and 4. A diameter followed by an H -block gives at most 10 sides in the active Eisenstein case, 5 in the Gaussian case, and 4 otherwise. Every major-arc bound is therefore below the corresponding minor-arc maximum.

This completes the proof of Theorem 1.2 for odd p, q .

4.5. The even semiprime case. It remains to prove $a(2p) = 6$. Let p be odd. The same residual-norm argument leaves the finite list in Appendix A.1. The classes $d = 1$ and $d = 3$ are elementary. For the other ten classes, the relation pattern depends only on the ideal class of a prime ideal above p , up to conjugation.

Lemma 4.7 (Prime-replacement invariance). *Fix an imaginary quadratic field whose unit group is $\{\pm 1\}$, and fix all ideal factors supported above 2 and the ramified primes. Let P, P' be unramified split prime ideals with $[P'] = [P]$. Then the integer relation lattice among the corresponding side angles, modulo rational multiples of π , is the same for P and P' . The same holds for signed major-arc relations.*

Proof. There is $\lambda \in K^\times$ with $P' = (\lambda)P$. If an admissible side type t has ideal factor $A_t P^{e_t} \overline{P}^{f_t}$, then replacing P by P' changes its phase by

$$(e_t - f_t) \operatorname{Arg}(\lambda) \pmod{\pi}.$$

For a signed relation with coefficients m_t , the total change is

$$\operatorname{Arg}(\lambda) \sum_t m_t (e_t - f_t).$$

The sum is exactly the coefficient of the independent p -basis angle and vanishes by Lemma 3.1. The converse follows by using λ^{-1} . Conjugate ideal classes negate all variable components and preserve the relation lattice. $\square$

Reduced positive binary quadratic forms represent ideal classes; see [2, Chapters 2–3]. Therefore one represented unramified prime per reduced-form orbit under conjugation suffices. The finite ideal-class verification proceeds as follows:

- (a) enumerate all 37 reduced-form orbits in the ten classes;
- (b) choose one represented unramified prime for each orbit;
- (c) enumerate all exact solutions of $s^2 + du^2 = 16p^2$;
- (d) determine the largest π -block, $\pi/2$ -block, and major-arc relation by exact multiplication in the biquadratic field $\mathbb{Q}(\sqrt{-d}, i) = \mathbb{Q}(\sqrt{d}, i)$;
- (e) separately check all 19 ramified pairs $p \mid d$ and the special $d = 3$ cases.

The exact computation finds no violation of

$$(31) \quad L_\pi \leq 6, \quad L_{\pi/2} \leq 3, \quad L_{\text{major}} \leq 4.$$

The regular hexagon gives equality. The remaining case $p = 2$, namely $n = 4$, is checked directly. Thus $a(2p) = 6$ for every prime p .

5. GROWTH ALONG FIXED POWERS

The semiprime theorem might suggest that few prime divisors force bounded behavior. The next construction shows that exponents alone can have the opposite effect. Here is a concrete preview: at radius $8192 = 2^{13}$, two integer chord lengths already form a 38-gon. The general argument explains why this phenomenon occurs for every fixed prime support.

Example 5.1 (A thirty-eight-gon at radius 8192). Take $N = 2$. Then $d = 15$, $v = 1$, and $z = 1 + \sqrt{-15}$. For

$$\tau = \frac{2 \arg z}{\pi}$$

one has the odd–odd Farey edge

$$\frac{5}{6} < \tau < \frac{11}{13}, \quad 6 \cdot 11 - 5 \cdot 13 = 1.$$

The scaled power coefficients are

$$\frac{\Re(z^6)}{2^5} = -7, \quad \frac{\Re(z^{13})}{2^{12}} = -2339.$$

At the common radius $2^{13} = 8192$, the first chord scales by 2^7 and has length 896, while the second has length 2339. Thus

$$26\alpha_{8192}(896) + 12\alpha_{8192}(2339) = \pi,$$

so

$$\{896^{[26]}, 2339^{[12]}\}$$

is a thirty-eight-gon and $a(8192) \geq 38$.

For this example, $\tau = 2 \arg(1 + \sqrt{-15})/\pi$ lies between the Farey neighbors $5/6$ and $11/13$. Figure 3 shows both the resulting polygon and the exact angle cancellation behind its closure.

Fix $N \geq 2$ and write

$$(32) \quad 4N^2 - 1 = dv^2,$$

where d is squarefree and $v \geq 1$. Put

$$z = 1 + v\sqrt{-d}, \quad N(z) = 4N^2,$$

and write

$$z^j = x_j + y_j\sqrt{-d} \quad (j \geq 1).$$

Lemma 5.2 (The quadratic chord tower). *For every $j \geq 1$, both x_j and y_j are divisible by 2^{j-1} . If*

$$X_j = \frac{x_j}{2^{j-1}}, \quad Y_j = \frac{y_j}{2^{j-1}},$$

then

$$(33) \quad X_j^2 + dY_j^2 = 4N^{2j}.$$

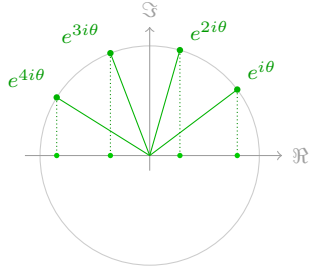
In particular, $|X_j|$ is an integer chord length at radius N^j . Moreover, if

$$\theta = \arg z \in (0, \pi/2), \quad \tau = \frac{2\theta}{\pi},$$

then τ is irrational and the associated half-angle is

$$(34) \quad \alpha_j := \alpha_{N^j}(|X_j|) = \arcsin |\cos(j\theta)|.$$

Figure 9 visualizes the angle supply created by the powers z^j : the normalized powers move around the unit circle, and their horizontal projections determine the integer chord lengths.



Since $z/(2N) = e^{i\theta}$,

$$\frac{z^j}{(2N)^j} = e^{ij\theta}, \quad \frac{X_j}{2N^j} = \cos(j\theta).$$

Thus each power supplies an integer chord $|X_j|$ with half-angle

$$\alpha_j = \arcsin |\cos(j\theta)|.$$

Because θ/π is irrational, the folded angles do not repeat periodically.

FIGURE 9. The quadratic chord tower, shown schematically with $\theta = 37^\circ$. Each normalized power gives another point on the unit circle; its real projection encodes the chord length and folded half-angle.

Proof. The real parts satisfy the trace recursion

$$x_{j+2} = 2x_{j+1} - 4N^2x_j, \quad x_0 = x_1 = 1.$$

With $X_j = x_j/2^{j-1}$ this becomes

$$(35) \quad X_{j+2} = X_{j+1} - N^2X_j, \quad X_1 = 1, \quad X_2 = 1 - 2N^2,$$

so all X_j are integers. The imaginary parts satisfy the same trace recursion, with $y_1 = v$ and $y_2 = 2v$; after division by 2^{j-1} the same integral recurrence applies. Hence $2^{j-1} \mid y_j$. Dividing $N(z^j) = (4N^2)^j$ by 4^{j-1} proves (33).

Since $z/(2N) = e^{i\theta}$,

$$\frac{X_j}{2N^j} = \cos(j\theta),$$

which gives (34). If θ/π were rational, then $e^{i\theta}$ would be a root of unity and

$$2 \cos \theta = \frac{1}{N}$$

would be a rational algebraic integer. It would therefore be an integer, which is impossible for $N \geq 2$. Hence θ/π and τ are irrational. Consequently $X_j \neq 0$, so this is a positive integer chord length. $\square$

For $e \geq j$, scaling the chord in Lemma 5.2 gives the integer side

$$(36) \quad s_{e,j} = N^{e-j} |X_j|$$

at radius N^e , with the same half-angle α_j .

We use one elementary fact about the Stern–Brocot path of an irrational number in $(0, 1)$. This is the Farey cancellation mechanism: an odd–odd bracketing edge makes two folded angles cancel exactly with integer multiplicities.

Lemma 5.3 (Odd–odd bracketing edges). *Let $\tau \in (0, 1)$ be irrational. Along its Stern–Brocot path there are infinitely many bracketing edges*

$$(37) \quad \frac{a_1}{b_1} < \tau < \frac{a_2}{b_2}, \quad b_1 a_2 - a_1 b_2 = 1,$$

for which both numerators a_1 and a_2 are odd. The denominators of these edges are unbounded.

Proof. At every Stern–Brocot step the two endpoint numerators cannot both be even, because the determinant is 1. If they have different parity, their median has odd numerator. Replacing the even endpoint therefore produces an odd–odd edge. If this never happened again, the algorithm would have to replace the odd endpoint at every subsequent step while keeping the even endpoint fixed. The successive medians would then converge to that fixed rational endpoint, forcing τ to equal it. This contradicts irrationality. After an odd–odd edge the median has even numerator, so the same argument restarts. Thus odd–odd edges occur infinitely often. Their denominators are unbounded because the Stern–Brocot intervals containing an irrational point form a strictly nested infinite sequence. $\square$

Proof of Theorem 1.3. Choose a sufficiently late odd–odd edge (37), so that $b_1, b_2 > 1$. Since its endpoints are Farey neighbors,

$$0 < b_1 \tau - a_1 < \frac{1}{b_2} < 1, \quad 0 < a_2 - b_2 \tau < \frac{1}{b_1} < 1.$$

Since a_1 and a_2 are odd, these are precisely the distances of $b_1 \tau$ and $b_2 \tau$ from the nearest odd integers. Formula (34) therefore gives, with $H = \pi/2$,

$$\alpha_{b_1} = H(b_1 \tau - a_1), \quad \alpha_{b_2} = H(a_2 - b_2 \tau).$$

The determinant identity now yields

$$(38) \quad b_2 \alpha_{b_1} + b_1 \alpha_{b_2} = H(b_1 a_2 - a_1 b_2) = H.$$

Doubling (38) gives a π -block with $2(b_1 + b_2)$ sides. By (36), both required chord lengths are integers at the common radius N^e , where $e = \max\{b_1, b_2\}$. Hence

$$(39) \quad a(N^e) \geq 2(b_1 + b_2) \geq 2e.$$

The odd-odd edges have unbounded denominators, so (5) follows. By Proposition 2.2, the sequence $a(N^e)$ is nondecreasing in e . Since the construction above makes it unbounded along an infinite subsequence, monotonicity implies that for every M there is an e_0 such that $a(N^e) \geq M$ for all $e \geq e_0$. Hence $a(N^e) \rightarrow \infty$. $\square$

6. PARITY WITH ONE GAUSSIAN SPLIT PRIME

The growth theorem is compatible with a striking parity constraint. We first separate the global geometric issue from the one-dimensional Gaussian optimization.

Put $H = \pi/2$. A *Gaussian H -block* is a multiset of non-diameter sides from the squarefree class $d = 1$ whose half-angles sum to H ; a *Gaussian π -block* is defined similarly. Let $H_G(n)$ and $P_G(n)$ be their maximum cardinalities, with value 0 if the corresponding class is absent.

Lemma 6.1 (Reduction of an odd maximum). *If $a(n)$ is odd, then an extremal polygon is a pure Gaussian minor-arc polygon. In particular, its side count is at most $P_G(n)$.*

Proof. By Theorem 3.3, an odd polygon must involve the Gaussian class. Consider first a minor-arc π -block. The splitting theorem makes the Gaussian subtotal a multiple of H . If other squarefree classes occur, positivity forces the Gaussian and non-Gaussian subtotals both to equal H . Let their side counts be g and r . Doubling either H -block gives an even minor-arc polygon with $2g$ or $2r$ sides. Since $g + r$ is odd, $g \neq r$, and

$$\max\{2g, 2r\} > g + r.$$

Thus a mixed odd minor-arc polygon cannot be extremal. An odd extremal minor-arc polygon must therefore be purely Gaussian.

If a diameter occurs, the remaining sides form an H -block with, say, $r \geq 2$ sides. Doubling that block gives a $2r$ -gon, which is larger than the original $(r + 1)$ -gon.

Finally consider a non-diameter major-arc relation

$$\alpha_* = \sum_j \alpha_j < H.$$

If the target is not Gaussian, the Gaussian restriction is a multiple of H lying in $(-H, 0]$, hence it is zero; no Gaussian source side occurs, and the parity statement for the classes $d > 1$ makes the total number of sides even. If the target is Gaussian, its Gaussian restriction lies in $[0, H)$ and must again be zero. Thus every source side is Gaussian and the relation is pure Gaussian. Writing $4n^2 - s_*^2 = u_*^2$, the complementary chord u_* is an integer and has half-angle $H - \alpha_*$. The source sides together with this complementary chord form an H -block with as many sides as the original major-arc polygon; doubling it gives a strictly larger even minor-arc polygon. Hence an odd maximum can only be a pure Gaussian π -block. $\square$

Assume now that n has exactly one distinct prime divisor $q \equiv 1 \pmod{4}$, and write $q^e \parallel n$. Choose a Gaussian prime

$$\varpi = A + iB, \quad q = \varpi \overline{\varpi}, \quad 0 < B < A,$$

and put

$$(40) \quad \theta = \arg \varpi, \quad \tau = \frac{4\theta}{\pi} \in (0, 1).$$

The number τ is irrational. Indeed, a rational value would make $\varpi/\overline{\varpi} = e^{2i\theta}$ a root of unity in $\mathbb{Q}(i)$; since $0 < \theta < \pi/4$, this is impossible.

For $1 \leq j \leq e$, let

$$\ell_j = \lfloor j\tau \rfloor, \quad r_j = j\tau - \ell_j.$$

Lemma 6.2 (The rank-one Gaussian angle set). *The non-diameter Gaussian half-angles at radius n are exactly*

$$(41) \quad Hr_j, \quad H(1 - r_j), \quad 1 \leq j \leq e.$$

Proof. Let $s + iu$ be a Gaussian integer with $s^2 + u^2 = (2n)^2$. A prime $r \equiv 3 \pmod{4}$ is inert in $\mathbb{Z}[i]$, so its exponent in $s + iu$ is forced and contributes no argument. The prime 2 is ramified and contributes only a multiple of H , which can be absorbed into the unit. For the unique split prime $q = \varpi\overline{\varpi}$, write the exponents of ϖ and $\overline{\varpi}$ in $s + iu$ as a and $2e - a$. Their contribution to the argument is

$$(a - (2e - a))\theta = 2(a - e)\theta.$$

Thus, modulo H , every nontrivial Gaussian direction is represented by $2j\theta = Hj\tau$ with $1 \leq j \leq e$, up to sign. Folding into $(0, H)$ gives Hr_j and $H(1 - r_j)$. Conversely, choosing exponents $e + j$ and $e - j$ (and an appropriate Gaussian unit) realizes both folded directions. Since τ is irrational, none of these angles is 0 or H . $\square$

The primes 2 and those congruent to 3 (mod 4) therefore change only the rational scale of the corresponding chords; they create no new Gaussian direction.

Let

$$(42) \quad \frac{a_1}{b_1} < \tau < \frac{a_2}{b_2}$$

be the two neighbors of τ in the Farey sequence of order e . Then

$$b_1a_2 - a_1b_2 = 1, \quad b_1, b_2 \leq e, \quad b_1 + b_2 > e.$$

Put

$$h = b_1 + b_2, \quad C = a_1 + a_2.$$

Lemma 6.3 (Exact Gaussian block maxima). *With the notation above,*

$$(43) \quad H_G(n) = h, \quad P_G(n) = 2h.$$

Proof. Represent the two angle types in (41) by

$$v_j = (-\ell_j, j), \quad v'_j = (\ell_j + 1, -j),$$

so that a vector (x, y) represents the angle $H(x + y\tau)$. An H -block therefore sums to $(1, 0)$ and a π -block to $(2, 0)$.

The median C/h lies strictly between the Farey neighbors (42). For every $1 \leq j \leq e$, the function $x \mapsto \lfloor jx \rfloor$ is constant on this Farey interval: a change would occur at a rational with denominator at most $j \leq e$, contradicting that the two endpoints are adjacent in the Farey sequence of order e . Moreover,

$\gcd(C, h) = 1$ by the determinant-one identity and $h > e \geq j$, so Cj/h is not an integer. Consequently

$$\ell_j = \left\lfloor \frac{Cj}{h} \right\rfloor,$$

and hence

$$(44) \quad 1 \leq Cj - h\ell_j \leq h - 1.$$

Define the integral linear functional

$$F(x, y) = hx + Cy.$$

By (44), every available vector v_j and v'_j has $1 \leq F \leq h - 1$. Thus an H -block has at most h sides and a π -block at most $2h$ sides.

For the lower bound, Farey adjacency gives

$$b_2(-a_1, b_1) + b_1(a_2, -b_2) = (1, 0).$$

Here $(-a_1, b_1) = v_{b_1}$ and $(a_2, -b_2) = v'_{b_2}$, because $a_1/b_1 < \tau < a_2/b_2$. Hence b_2 copies of the first angle and b_1 copies of the second form an H -block with $b_1 + b_2 = h$ sides. Doubling it gives a π -block with $2h$ sides. This proves (43). $\square$

Proof of Theorem 1.5. If $\omega_1(n) = 0$, Theorem 3.3 shows that every admissible polygon already has even side count. Suppose $\omega_1(n) = 1$. If $a(n)$ were odd, Lemma 6.1 would reduce an extremal polygon to a pure Gaussian π -block. But Lemma 6.3 shows that the largest such block has the even cardinality $P_G(n) = 2h$. This block is itself an admissible polygon, so $a(n) \geq P_G(n)$, while the reduction lemma gives $a(n) \leq P_G(n)$ if the maximum is odd. Thus an odd maximum would have to equal the even number $P_G(n)$, a contradiction. Therefore $a(n)$ is even. $\square$

The rank-one argument pairs angles along one ordered interval. With several split primes, the coefficient data occupy a higher-dimensional lattice; Figure 10 illustrates this contrast.

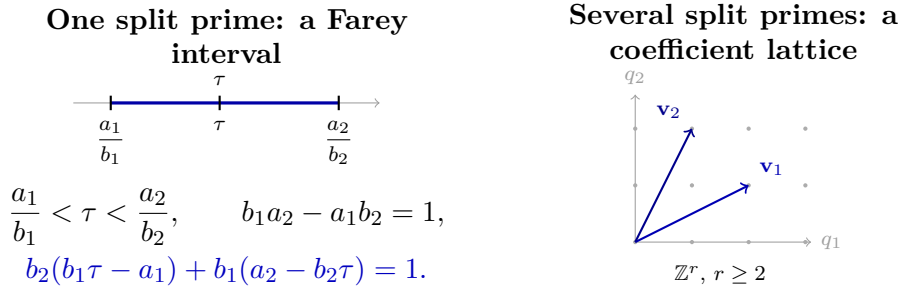


FIGURE 10. Why the parity proof is one-dimensional. With one split Gaussian prime, Farey neighbors bracket a single parameter and the colored weighted integer gaps sum to the Farey determinant 1. Several split primes create multiple coefficient coordinates; the schematic right panel depicts no particular zero-sum relation.

Remark 6.4. The proof identifies the precise point at which the one-prime hypothesis is used. With several distinct primes $q \equiv 1 \pmod{4}$, Gaussian angles are indexed by a multidimensional exponent box rather than an interval. The Farey dual functional is then replaced by a bounded zero-sum problem in a higher-dimensional lattice. The general conjecture that $a(n)$ is always even remains open.

7. EXAMPLES AND CONSEQUENCES

The three main results can be seen in small concrete cases.

Example 7.1 (Eisenstein jump). For $p = 7$ and $q = 13$,

$$(s_7, u_7) = (2, 8), \quad (s_{13}, u_{13}) = (1, 15).$$

Formula (26) gives $S = 61$. Thus

$$\{7^{[6]}, 26^{[6]}, 61^{[6]}\}$$

is the side multiset of an eighteen-gon of radius 91, and $a(91) = 18$.

Example 7.2 (Gaussian jump). For $p = 5$ and $q = 13$, the Gaussian construction gives

$$\{32^{[2]}, 50^{[4]}, 66^{[2]}\},$$

so $a(65) = 8$.

Corollary 7.3. *For a prime p ,*

$$a(p^2) = 18$$

exactly when $p \equiv 1 \pmod{3}$ and $2x_p < \pi/6$; otherwise $a(p^2) = 6$. In particular,

$$a(49) = 18, \quad a(169) = 18, \quad a(361) = 6.$$

On the other hand, Theorem 1.3 gives $a(p^e) \rightarrow \infty$ as $e \rightarrow \infty$, while Theorem 1.5 shows that every one of these maxima is even.

This contrast is the main point of the paper: the first two prime factors already determine the complete semiprime picture, but no fixed prime support bounds the number of sides once exponents are allowed to grow.

8. OUTLOOK

The semiprime classification is complete, and the parity theorem settles the Gaussian rank-one case. The natural global conjecture is

$$a(n) \equiv 0 \pmod{2} \quad (n \geq 1).$$

The evidence is structural: the theorem proves it whenever at most one distinct prime divisor $q \equiv 1 \pmod{4}$ occurs, and the computed initial values agree. In this rank-one case, the Gaussian component vectors lie on a line and Farey pairing forces an even side count. With several split primes, cancellations can involve several coordinates at once; the one-dimensional argument no longer controls the length of a zero-sum relation. A higher-dimensional parity invariant, or an odd counterexample, would settle the conjecture.

The other open direction is quantitative. Theorem 1.3 gives the universal lower bound $\limsup a(N^e)/e \geq 2$, but it does not determine the true growth

rate for any fixed N . Even the binary sequence $a(2^e)$ presents a concrete next problem: determine its order of growth, or sharpen the constant 2 in the limsup bound.

The semiprime problem is completely rigid, the fixed-support power sequences are unbounded, and global parity remains open. Understanding how these three behaviours coexist under the same exact angle-closure condition is a natural next step in the theory.

APPENDIX A. EXACT VERIFICATION AND REPRODUCIBILITY

A.1. Residual squarefree class lists. For distinct odd primes, the non-Gaussian classes arising from the residual norm equation are

$$(A.1) \quad d \in \{3, 7, 15, 23, 31, 39, 55, 87, 231, 247, 255\}.$$

For a repeated odd prime, the corresponding non-Gaussian list is

$$(A.2) \quad d \in \{3, 7, 15, 39, 55\}.$$

For the even semiprime case $n = 2p$, the full list, including the Gaussian and Eisenstein classes, is

$$(A.3) \quad d \in \{1, 3, 7, 15, 23, 31, 39, 55, 87, 231, 247, 255\}.$$

These lists arise from the primitive equations $x^2 + dy^2 = L^2$ and the bounds on L established in the main text.

A.2. The 37 reduced-form orbits and a sample probe. For each ordinary field in the even case, the table lists one reduced primitive positive binary quadratic form from each orbit under conjugation $[a, b, c] \mapsto [a, -b, c]$. The forms have discriminant $-d$ and satisfy the usual reduced conditions $-a < b \leq a \leq c$, with $b \geq 0$ on the boundary. The entries therefore make the orbit count used in the finite verification explicit; $\omega(d)$ is the number of ramified rational primes $p \mid d$.

TABLE 2. Reduced-form orbit representatives for the ten ordinary fields in the even semiprime verification.

d	Orbits	$\omega(d)$	Representatives $[a, b, c]$
7	1	1	(1, 1, 2)
15	2	2	(1, 1, 4), (2, 1, 2)
23	2	1	(1, 1, 6), (2, -1, 3)
31	2	1	(1, 1, 8), (2, -1, 4)
39	3	2	(1, 1, 10), (2, -1, 5), (3, 3, 4)
55	3	2	(1, 1, 14), (2, -1, 7), (4, 3, 4)
87	4	2	(1, 1, 22), (2, -1, 11), (3, 3, 8), (4, -3, 6) (1, 1, 58), (2, -1, 29), (3, 3, 20), (4, -3, 15),
231	8	3	(5, -3, 12), (6, -3, 10), (7, 7, 10), (8, 5, 8)
247	4	2	(1, 1, 62), (2, -1, 31), (4, -3, 16), (8, 3, 8) (1, 1, 64), (2, -1, 32), (3, 3, 22), (4, -1, 16),
255	8	3	(5, 5, 14), (6, -3, 11), (7, -5, 10), (8, 1, 8)
Total	37	19	

Here is one exact input to the even-case search. The field $d = 7$ has class number one, represented by $[1, 1, 2]$, and this form represents $p = 11$ since $1^2 + 1 \cdot 1 \cdot 2 + 2 \cdot 2^2 = 11$. At radius $n = 22$, the divisors of $2n = 44$ give the following complete list of positive primitive solutions to $A^2 + 7B^2 = M^2$:

$M \mid 44$	Primitive pairs (A, B)
1, 2, 22	none
4	(3, 1)
11	(3, 4)
44	(19, 15), (37, 9)

For each listed divisor, completeness can be checked using $0 \leq B \leq \lfloor M/\sqrt{7} \rfloor$, testing whether $M^2 - 7B^2$ is a square, and retaining only pairs with $\gcd(A, B) = 1$. Since $s = (44/M)A$, these correspond to the four side lengths 33, 12, 19, 37. With $\pi_2 = (1 + \sqrt{-7})/2$ and $\pi_{11} = 2 + \sqrt{-7}$, exact factorization of $A + B\sqrt{-7}$ gives the normalized signed exponent vectors $(v_2, v_{11}) = ((e_2 - f_2)/2, (e_{11} - f_{11})/2)$

$$(-1, 0), \quad (0, -1), \quad (1, 1), \quad (1, -1),$$

respectively, up to the unit ± 1 . This exhibits concretely how primitive norm solutions become the integer vectors whose cancellations the finite block search tests.

All finite checks use exact integer arithmetic and multiplication in the basis

$$1, \quad \sqrt{d}, \quad i, \quad i\sqrt{d}.$$

No trigonometric approximation enters these checks. In a fresh Maple 2026 replay on 2 October 2026, the odd structural certificate passed, and the even ideal-class audit returned 37 unramified reduced-form orbits and 19 ramified pairs with no violations. The independent recursion–Cartesian cross-audit found no mismatches and no bound violations. A separate Maple program exhausts multiplicity vectors under derived finite bounds; independent Python programs reproduce the candidate classes, orbit counts, and extremal bounds. Branches are pruned only when their positive partial angle sum exceeds the target, and strict monotonicity (8) identifies each target angle with a unique side length. The source code, recorded outputs, cross-checks, reproduction instructions, and SHA-256 manifest are supplied as Online Resource 1 ([ESM_1.zip](#)).

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Data and code availability. The first 400 terms of OEIS A398490 are used for the contextual plot in Figure 2. The exact Maple and Python source code, recorded outputs, cross-checks, and reproduction instructions supporting the computer-assisted semiprime verification are provided in the accompanying file `ESM_1.zip`. The DOI for this version is 10.5281/zenodo.23106967.

Use of artificial intelligence. OpenAI ChatGPT was used during August–October 2026 as an AI-assisted research and manuscript-preparation tool; the earlier proof-development and audit stages used GPT-5.6 Sol. Prompts asked for candidate proof strategies, adversarial checks of individual proof steps, and suggestions on organization and wording. An additional editorial pass in October focused on narrative transitions, figure captions, the Outlook, and presentation. The author decided which suggestions to pursue, independently reconstructed and verified every proof, calculation, and reference, and made all final judgments about the arguments, interpretation, and conclusions. No AI-generated output is used as mathematical evidence, and no research data were generated by AI. The figures are deterministic TikZ diagrams of the mathematical constructions. The author reviewed and approved the final text and takes full responsibility for its content.

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